Simulation Analysis of MPPT Control Strategy for Photovoltaic Cells Based on Perturbation Observation Method

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***Abstract*—Energy is the cornerstone of economic development in all countries. With the massive consumption of non-renewable energy sources such as coal and oil, the reserves of fossil energy in various countries have decreased sharply. It is now extremely urgent to vigorously develop new types of green and renewable energy. Photovoltaic power generation technology can convert solar energy into light energy. In order to improve the power generation efficiency of photovoltaic cells, this paper conducts a simulation analysis of the maximum power point tracking control strategy of photovoltaic cells based on the perturbation observation method. Firstly, a Simulink simulation model of photovoltaic cells was established based on their working principle, basic composition and mathematical model, and the response characteristics of photovoltaic cells were analyzed. Secondly, the adoption of MPPT (Maximum Power Point Tracking) technology can ensure that photovoltaic modules output electrical energy at the maximum power. The disturbance observation method, which is simple to control, easy to implement and highly practical, is adopted, and the disturbance observation method has been studied and analyzed. Finally, the model of the perturbation observation method was built on the Simulink platform and simulation analysis was carried out. It has been verified that the perturbation observation method used can track the maximum power point quickly and accurately.**

Keywords—Solar energy; Photovoltaic cell; Maximum power point tracking; Perturbation observation method

I. INTRODUCTION

To protect the environment and ensure energy supply, it is necessary to develop renewable and clean energy. Therefore, all countries have reached a consensus that while developing the economy, environmental protection should be achieved, the path of sustainable development should be followed, and the utilization of clean and renewable energy should be increased [1]. China is rich in new energy resources and has huge development potential, especially in solar energy.

In order to accelerate the development speed of photovoltaic cells, various countries have begun to optimize the various shortcomings of photovoltaic cells. At present, there are mainly two solutions: The first approach is to delve deeper into the development and search for new materials for photovoltaic cells, research and develop the physical materials currently used, and further enhance the photoelectric conversion efficiency of photovoltaic cells; The second method is to utilize MPPT technology and apply the MPPT controller to photovoltaic cells. The MPPT control algorithm dynamically adjusts the output power of the photovoltaic cells in real time to reach the position of the maximum power point, thereby improving the power generation efficiency of the photovoltaic cells and reducing power waste. As the development of new materials is still in the stage of in-depth learning and experimentation at present, the second approach is the main solution currently adopted. This method can significantly enhance the photoelectric conversion efficiency of photovoltaic cells at a relatively low cost, reduce energy loss of photovoltaic cells, and indirectly extend the service life of photovoltaic cells [2].

This paper first uses MATLAB/Simulink to model, simulate and analyze the characteristics of photovoltaic cells; Then the Boost boost circuit is designed to obtain a stable 750V DC bus voltage; Finally, an improved MPPT control algorithm is proposed. Through theoretical analysis and simulation verification, it is found that this algorithm can quickly track the maximum power point of the photovoltaic array, thereby achieving the output of the maximum power of the photovoltaic power generation system.

II. THE MODEL OUTPUT CHARACTERISTICS OF PHOTOVOLTAIC CELLS

A. Construction of Photovoltaic Cell Models

The output current of the volt cell varies with the change of light intensity.When the intensity of external light is constant, photovoltaic cells can be approximately regarded as a kind of constant The flow source, its equivalent circuit is shown in Figure 1.

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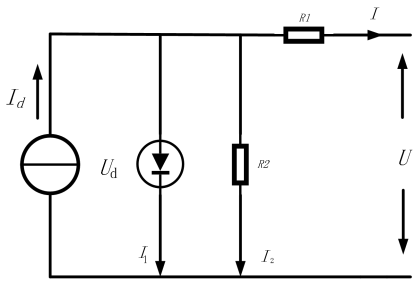


Fig. 1. Equivalent circuit of photovoltaic cells.

For the current of photovoltaic cells, it mainly depends on the characteristics of the semiconductor materials of the cells and is linearly related to the irradiance level and temperature. Then there is:

$$I = I_d - I_0\left\{\exp\left[\frac{q(U+IR_1)}{AkT}\right]-1\right\}-\frac{U+IR_1}{R_2} \quad (1)$$

It can be obtained by combining the above formulas:

$$I = I_d - I_0[\exp(\frac{qU}{36AKT})-1] \quad (2)$$

The Simulink simulation model of the above-mentioned equivalent circuit was built in MATLAB. Since the solution of parameters R_S and R_{sh} is relatively inconvenient in engineering applications, the model was improved. Photovoltaic module manufacturers usually provide the current I_m at the maximum power point, the voltage U_m at the maximum power point, the open-circuit voltage U_{OC}, and the short-circuit circuit current I_{SC} under standard test conditions. Therefore, the model can be constructed based on the above parameters, as shown in Figure 2.

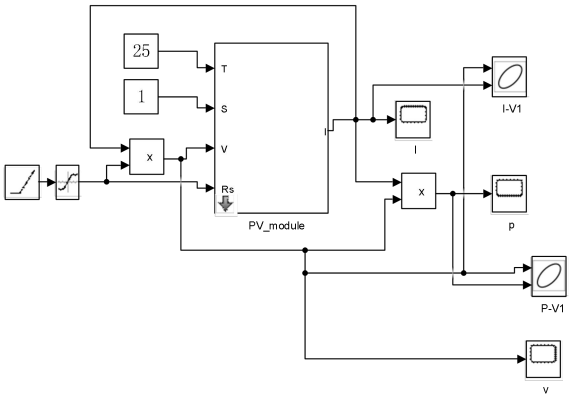
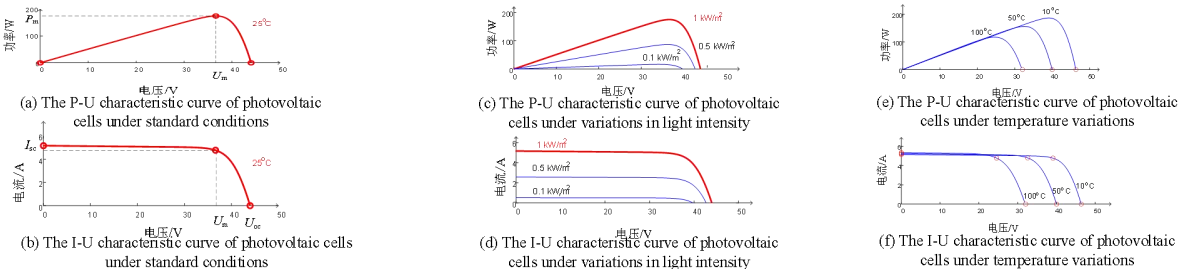


Fig. 2. Photovoltaic cell model.

B. *Analysis of Photovoltaic Cell Characteristics*

This paper takes the photovoltaic modules selected for a certain photovoltaic power generation system as the research object and obtains the P-U and I-U curves under standard conditions (light intensity 1000W/㎡, 25℃), as shown in Figure 3. As can be seen from Figure 3, the maximum output power of this photovoltaic cell under standard working conditions is 307.371W, the output voltage is 29.692V, the short-circuit current is 10.78A, and the open-circuit voltage is 40.026V, which are basically consistent with the parameters in the photovoltaic module technical parameter table..



(a) The P-U characteristic curve of photovoltaic cells under standard conditions

(b) The I-U characteristic curve of photovoltaic cells under standard conditions

(c) The P-U characteristic curve of photovoltaic cells under variations in light intensity

(d) The I-U characteristic curve of photovoltaic cells under variations in light intensity

(e) The P-U characteristic curve of photovoltaic cells under temperature variations

(f) The I-U characteristic curve of photovoltaic cells under temperature variations

Fig. 3. Photovoltaic cell P-U and I-U characteristic curves.

The output characteristics of photovoltaic cells under standard temperature of 25℃, different light conditions and standard light intensities, and different temperatures were further analyzed. The results are shown in Figure 3.

As can be seen from Figure 3, when the ambient temperature of the photovoltaic cell remains constant and the light intensity gradually increases, its open-circuit voltage gradually rises and the short-circuit current gradually increases [3]. From this, it can be known that the incident irradiance is positively correlated with the short-circuit current. When the ambient temperature in which the battery is located is relatively stable, the stronger the incident irradiance, the greater the maximum product of the current and voltage, that is, the greater the maximum output power of the component.When photovoltaic cells are exposed to stable light, as the ambient temperature rises, the open-circuit voltage of the photovoltaic cells gradually decreases. The short-circuit current varies little with temperature. Therefore, temperature has a significant impact on the open-circuit voltage of photovoltaic cells, but a relatively small influence on the short-circuit current. Under constant light conditions, changes in environmental temperature have a certain impact on the output power of photovoltaic cells, but the degree of influence is relatively small.

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III. Design and Simulation Analysis of Photovoltaic MPPT Control Algorithm

A. MPPT control Principle and commonly used control algorithms

The output of photovoltaic arrays has nonlinear characteristics and is affected by factors such as light intensity, ambient temperature and load. When the output voltage reaches a certain value, the output power of the photovoltaic array can reach its maximum value, and this point is called the maximum power point. To enhance the utilization rate of photovoltaic panels, it is essential to keep the system operating near the maximum power point. Among various algorithms, the perturbation observation method is widely used in the maximum power point tracking of photovoltaic arrays due to its simple structure. In this paper, an improved variable step size perturbation observation method is adopted to verify the tracking effect of the maximum power point [4].

B. The workflow of the disturbance observation method

When the perturbation observation method is adopted, the smaller the step size, the smaller the search range near the maximum power point of the photovoltaic array, the smaller the output power loss, but the longer the tracking time. When the step size is larger, the tracking speed is fast, but the tracking accuracy is not high enough. The search range near the maximum power point is large, and the output power fluctuates greatly. Therefore, tracking accuracy and response speed are a pair of negatively correlated quantities [5]. However, for a fixed step size, its tracking accuracy and time need to meet the design requirements through step size Settings, which often consumes a lot of effort. Therefore, the step size needs to be changed.

The variable-step perturbation observation method can automatically change the step size based on the rate of change of the power point. Mainly through the slope change in the U-P characteristic curve, the slope change is 0 at the maximum power point, and the slope change is relatively slow near the maximum power point. At this time, the disturbance step size needs to be reduced to decrease the power fluctuation range and better achieve the tracking of the maximum power. The slope changes relatively rapidly at the point far from the maximum power. At this time, it is necessary to increase the disturbance step size and expand the power fluctuation range, so as to approach the maximum power point more quickly.

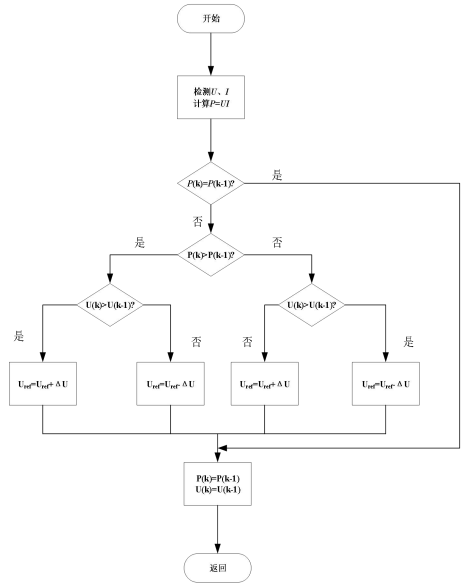


Fig. 4. Flowchart of the disturbance observation method.

C. Simulation analysis

The simulation model of MPPT was built in Simulink, as shown in Figure 5. The research object of this paper is a certain photovoltaic power generation system. The frequency is set at 10 KHz, the voltage ripple is 0.1%, the current ripple is 2.5%, the isolation capacitor is e^{-6}F, the resistance is $e^{-6}\,\Omega$, and the moment factor g is taken as 1.

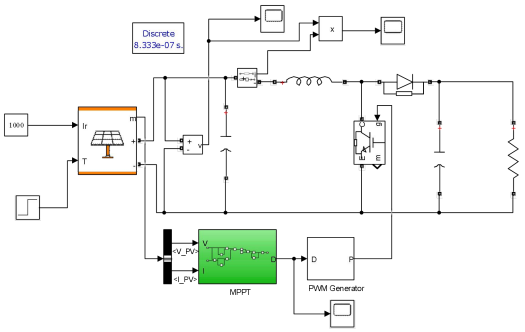


Fig. 5. The Simulink simulation model of MPPT.

TABLE I. *Parameters of Photovoltaic Cell and Circuit Simulation Model*

Photovoltaic cell parameters		*Circuit parameters*	
$U_{oc}(V)$	36.3	$I_{mp}(A)$	7.35
$I_{sc}(A)$	7.84	$R_{sh}(\Omega)$	313.0553
$U_{mp}(V)$	29	$R_s(\Omega)$	0.39381
$P_{max}(\Omega)$	213.15	$I_L(A)$	7.8654
PWM (Hz)	12000		

Because photovoltaic cells are greatly affected by light, the simulation experiment mainly uses changes in light to simulate weather conditions. To verify the effect of the system's maximum power point tracking control when the

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external environment changes, the following simulation is conducted：

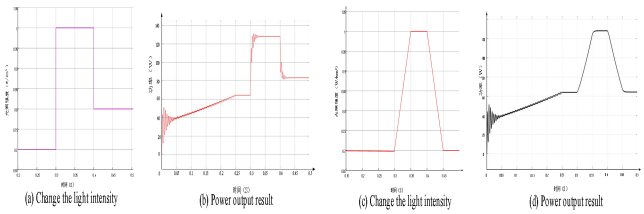


Fig. 6. Simulation analysis when the light intensity is changed

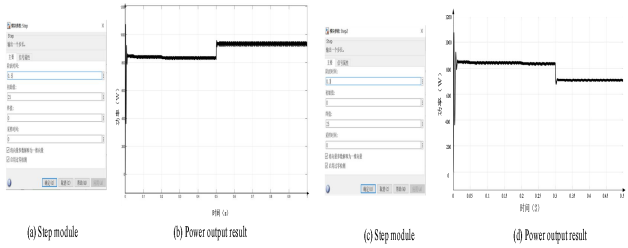


Fig. 7. Simulation analysis when temperature is changed

The power waveforms of the simulation results are shown in Figures 6 and 7. At 0s, the system starts up. Under the initial setting conditions of a temperature of 25 ℃ and a light intensity of 1000, the disturbance observation method tracks to the maximum power point after 0.02 seconds. When the light intensity remains constant, at 0.5 seconds, when the temperature is changed from the initial 25℃ to 0℃, as the temperature changes, the output power of the photovoltaic cell will also change accordingly. The disturbance observation method will re-track. Under the control of MPPT, the maximum power point of the photovoltaic cell will be tracked at 0.531 seconds. The output power has changed from 840W to approximately 917.3W.

The system started at 0s. Under the initial setting conditions of a temperature of 25℃ and a light intensity of 1000 W/m2, the disturbance observation method tracked to the maximum power point after 0.015 seconds. When the light intensity remains constant, at 0.3 seconds, when the temperature is changed from the initial 25℃ to 50℃, as the temperature changes, the output power of the photovoltaic cell will also change accordingly. The disturbance observation method will re-track. Under the control of MPPT, the maximum power point of the photovoltaic cell will be tracked at 0.307 seconds. The output power has changed from 851.8W to approximately 705.9W. That is, when the system operates for 0.3 seconds, the temperature rises, but the output power curve of the photovoltaic cell drops, and the system can quickly maintain at the maximum power output state. However, a misjudgment occurred at 0.3 seconds, but it quickly returned to the maximum power point and performed correct power tracking.

From the above simulation analysis diagram, it can be concluded that the photovoltaic cell adopting the disturbance observation method can track the maximum power point in an extremely short time. However, when tracking near the maximum power point, it will still fluctuate continuously, that is, the operating point of the photovoltaic cell will inevitably oscillate near the maximum power point.

IV. CONCLUSIONS

The output characteristics of solar photovoltaic cells are influenced by various factors such as light intensity, ambient temperature and load. To enhance the output power of photovoltaic panels, it is necessary to track the maximum power point in real time. In this paper, the characteristics of photovoltaic cells, boost circuits and MPPT algorithms are studied by using MATLAB/Simulink.

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